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User's Manual
TEVA-SPOT Toolkit 2.3

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Forward

Since its inception in 1970, EPA's mission has been to pursue a cleaner, healthier environment for the American people. The Agency was assigned the daunting task of repairing the damage already done to the natural environment and establishing new criteria to guide Americans in making a cleaner environment a reality. Since 1970, the EPA has worked with federal, state, tribal, and local partners to advance its mission to protect human health and the environment. In order to carry out its mission, EPA employs and collaborates with some of the nation's best scientific minds. EPA prides itself in applying sound science and state of the art techniques and methods to develop and test innovations that will protect both human health and the environment.

Under existing laws and recent Homeland Security Presidential Directives, EPA has been called upon to play a vital role in helping to secure the nation against foreign and domestic enemies. The National Homeland Security Research Center (NHSRC) was formed in 2002 to conduct research in support of EPA's role in homeland security. NHSRC research efforts focus on five areas: water infrastructure protection, threat and consequence assessment, decontamination and consequence management, response capability enhancement, and homeland security technology testing and evaluation. EPA is the lead federal agency for drinking water and wastewater systems and the NHSRC is working to reduce system vulnerabilities, prevent and prepare for terrorist attacks, minimize public health impacts and infrastructure damage, and enhance recovery efforts.

This Users Manual for the TEVA-SPOT Toolkit software package is published and made available by EPA's Office of Research and Development to assist the user community and to link researchers with their clients.

Jonathan Herrmann, Director

National Homeland Security Research Center
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- **PICO** - BSD and Common Public License
- **randomsample**, **sideconstraints** - ATT Software for noncommercial use.
- **ufl** - Common Public License

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1 Introduction

Public water distribution systems are inherently vulnerable to accidental or intentional contamination because of their distributed geography. Further, there are many challenges to detecting contaminants in drinking water systems: municipal distribution systems are large, consisting of hundreds or thousands of miles of pipe; flow patterns are driven by time-varying demands placed on the system by customers; and distribution systems are looped, resulting in mixing and dilution of contaminants. The use of on-line, real-time contaminant warning systems (CWSs) is a promising strategy for mitigating these risks. Online sensor data can be combined with public health surveillance systems, physical security monitoring, customer complaint surveillance, and routine sampling programs to effect a rapid response to contamination incidents (**?**).

A variety of technical challenges need to be addressed to make CWSs a practical, reliable element of water security systems. A key aspect of CWS design is the strategic placement of sensors throughout the distribution network. Given a limited number of sensors, a desirable sensor placement minimizes the potential economic and public health impacts of a contaminant incident. There are a wide range of important design objectives for sensor placements (e.g., minimizing the cost of sensor installation and maintenance, the response time to a contamination incident, and the extent of contamination). In addition, flexible sensor placement tools are needed to analyze CWS designs in large scale networks.

1.1 What is TEVA-SPOT?

The Threat Ensemble Vulnerability Assessment and Sensor Placement Optimization Tool (TEVA-SPOT) has been developed by the U. S. Environmental Protection Agency, Sandia National Laboratories, Argonne National Laboratory, and the University of Cincinnati. TEVA-SPOT has been used to develop sensor network designs for several large water utilities **?**, including the pilot study for the EPA's Water Security Initiative.

TEVA-SPOT allows a user to specify a wide range of modeling inputs and performance objectives for contamination warning system design. Further, TEVA-SPOT supports a flexible decision framework for sensor placement that involves two major steps: a modeling process and a decision-making process **?**. The modeling process includes (1) describing sensor characteristics, (2) defining the design basis threat, (3) selecting impact measures for the CWS, (4) planning utility response to sensor detection, and (5) identifying feasible sensor locations.

The design basis threat for a CWS is the ensemble of contamination incidents that a CWS should be designed to protect against. In the simplest case, a design basis threat is a contamination scenario with a single contaminant that is introduced at a specific time and place. Thus, a design basis threat consists of a set of contamination incidents that can be simulated with standard water distribution models **?**. TEVA-SPOT provides a convenient interface for defining and computing the impacts of design basis threats. In particular, TEVA-SPOT can simulate many contamination incidents in parallel, which has reduced the computation of very large design basis threats from weeks to hours on the EPAs high performance computing system.

TEVA-SPOT was designed to model a wide range of sensor placement problems. For example, TEVA-SPOT supports a number of impact measures, including the number of people exposed to dangerous levels of a contaminant, the volume of contaminated water used by customers, the number of feet of contaminated pipe, and the time to detection. Response delays can also be specified to account for the time a water utility would need to verify a contamination incident before notifying the public. Finally, the user can specify the feasible locations for sensors and fix sensor locations during optimization. This flexibility allows a user to evaluate how different factors impact the CWS performance and to iteratively refine a CWS design.